

MIT OpenCourseWare

7.06

Cell Biology 7.06 — Cell Biology May 2026 **EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (5 points) Prove the fundamental theorem concerning Membrane Transport in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
2. (5 points) Analyze the limitations of standard approaches to Cytoskeleton when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (4 points) Analyze the limitations of standard approaches to Cell Cycle when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
4. (3 points) Prove the fundamental theorem concerning Apoptosis in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (4 points) Analyze the limitations of standard approaches to Cell Signaling when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
6. (5 points) Analyze the limitations of standard approaches to Membrane Transport when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (8 points) Analyze the limitations of standard approaches to Cytoskeleton when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
8. (2 points) Prove the fundamental theorem concerning Cell Cycle in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
9. (4 points) Prove the fundamental theorem concerning Apoptosis in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

10. (4 points) Construct an original proof-level problem involving Cell Signaling for 7.06 (Cell Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
11. (2 points) Prove the fundamental theorem concerning Membrane Transport in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (4 points) Construct an original proof-level problem involving Cytoskeleton for 7.06 (Cell Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
13. (5 points) Analyze the limitations of standard approaches to Cell Cycle when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (2 points) Analyze the limitations of standard approaches to Apoptosis when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
15. (2 points) Construct an original proof-level problem involving Cell Signaling for 7.06 (Cell Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
16. (3 points) Prove the fundamental theorem concerning Membrane Transport in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
17. (5 points) Analyze the limitations of standard approaches to Cytoskeleton when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
18. (4 points) Prove the fundamental theorem concerning Cell Cycle in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (4 points) Analyze the limitations of standard approaches to Apoptosis when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.
20. (5 points) Analyze the limitations of standard approaches to Cell Signaling when applied to edge cases in 7.06. Construct explicit counterexamples showing where intuitive reasoning fails.

21. (7 points) Construct an original proof-level problem involving Membrane Transport for 7.06 (Cell Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (3 points) Prove the fundamental theorem concerning Cytoskeleton in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Prove the fundamental theorem concerning Cell Cycle in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (5 points) Prove the fundamental theorem concerning Apoptosis in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
25. (3 points) Prove the fundamental theorem concerning Cell Signaling in the context of 7.06 (Cell Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cytoskeleton. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cell Cycle. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 4:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cell Signaling. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Membrane Transport. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cytoskeleton. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 8:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 10:

Solution approach: First recall the definitions and key theorems related to Cell Signaling from 7.06. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 11:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

Solution approach: First recall the definitions and key theorems related to Cytoskeleton from 7.06. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 13:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cell Cycle. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 14:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Apoptosis. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:

Solution approach: First recall the definitions and key theorems related to Cell Signaling from 7.06. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 16:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 17:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cytoskeleton. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 18:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Apoptosis. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cell Signaling. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Membrane Transport from 7.06. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 22:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 25:

The proof follows from the definitions and properties established in 7.06. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.